

Assignment on Multiprocessor Computing Systems: Development of an application on the ARM NEON coprocessor of Xilinx Zynq-7000

December 16, 2024

Working Group

	Name and Surname	Registration Number	Email or phone
1.	Francisco Lara Santiago	ERASMUS	franlasant@correo.ugr.es
2.	Francisco Javier Checa Casas	ERASMUS	javicheca2003@gmail.com

1. Introduction

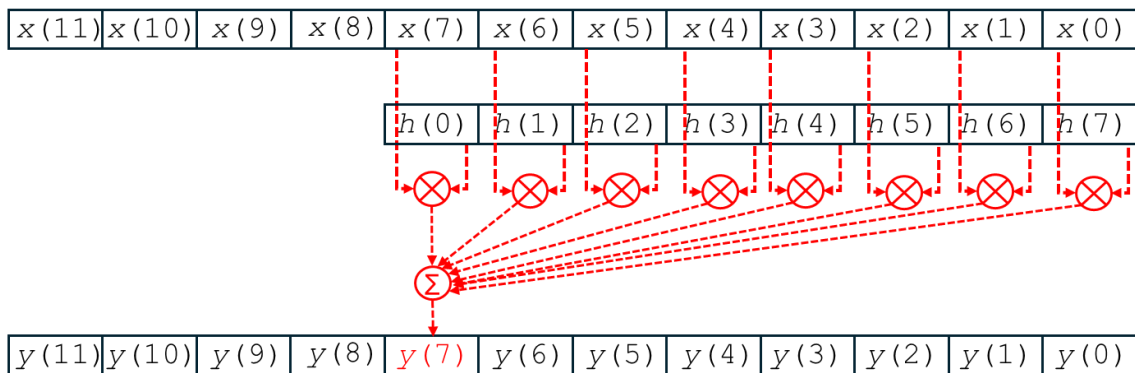
The assignment involves implementing a simple C program that will run on the NEON coprocessor of the ARM processor on the Xilinx Zynq-7000 chip, which is located on the Zybo board. The purpose of the assignment is to work on the development of a simple application in a processing system that offers parallel processing capabilities of the SIMD (Single Instruction Multiple Data) category.

2. Brief System Description

The system must implement a digital Finite Impulse Response (FIR) filter. An FIR filter accepts as input a sequence of samples $x(n)$ (discrete-time signal) and derives as output another sequence of samples $y(n)$, where $n \in \mathbb{Z}$. The input-output relationship is defined by the convolution equation.

$$y(n) = \sum_{k=0}^{N-1} h(k)x(n-k)$$

The sequence $h(k)$, $k=0 \dots N-1$ is called the impulse response of the filter, and the integer N is the order of the filter. The behavior of the filter is determined solely by its impulse response. The computation of the output sample $y(7)$ for a filter of order $N=8$ is illustrated in the following diagram:



Implement two applications, one for serial computation of the convolution and one for parallel computation using NEON's Q registers. For both applications use single floating point arithmetic (C types `float` and `float32x4_t`, respectively). For the filter's impulse response $h(k)$, use the following values (filter order $N=8$):

$h(0)$	$h(1)$	$h(2)$	$h(3)$	$h(4)$	$h(5)$	$h(6)$	$h(7)$
-0.0010	0.1219	0.1862	0.2429	0.2429	0.1862	0.1219	-0.0010

As input, use a cosine signal of the form $x(n)=\cos(\omega n)$, $n=0 \dots 31$, with amplitude 1, zero initial phase, and circular (or angular) frequency ω . For the circular frequency ω , experiment with two values: $\omega = \omega_1 = 2\pi/16$ and $\omega = \omega_2 = 2\pi/8$. (Period $T_1 = 16$ samples and $T_2 = 8$ samples, respectively).

Each program must calculate the filter output $y(n)$ for $n=7 \dots 31$.

Calculate the speedup and efficiency of the parallel algorithm compared to the corresponding serial one. In the time measurements, consider only the convolution computation and exclude the memory preparation times for the input signal $x(n)$ and the display times (in ASCII format) for the input signal $x(n)$ and the output signal $y(n)$ on the terminal.

3. Observations

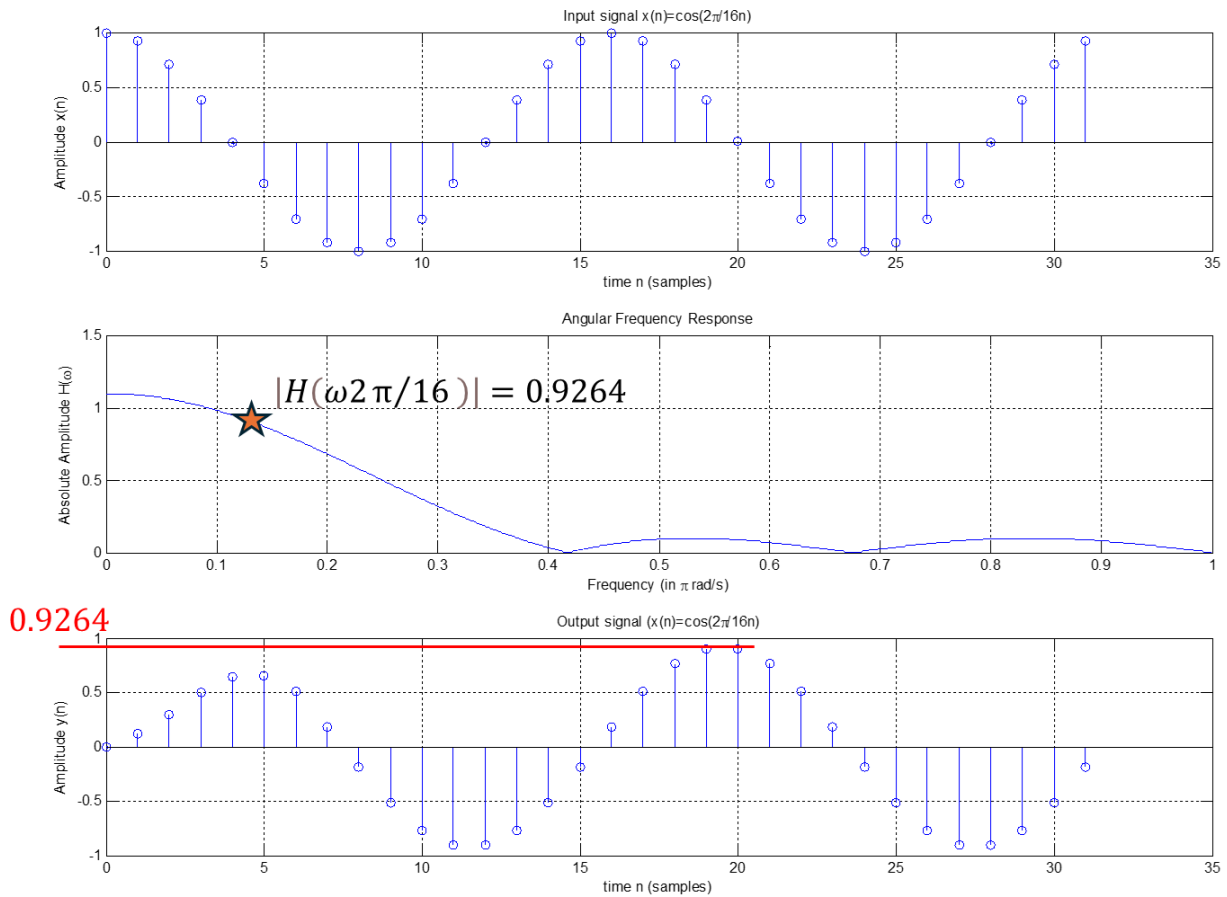
A key property of an FIR filter is that its excitation with a cosine (or sine) signal results in an output response that is a cosine (or sine) signal of exactly the same circular frequency ω , but with a different amplitude and phase. Namely,

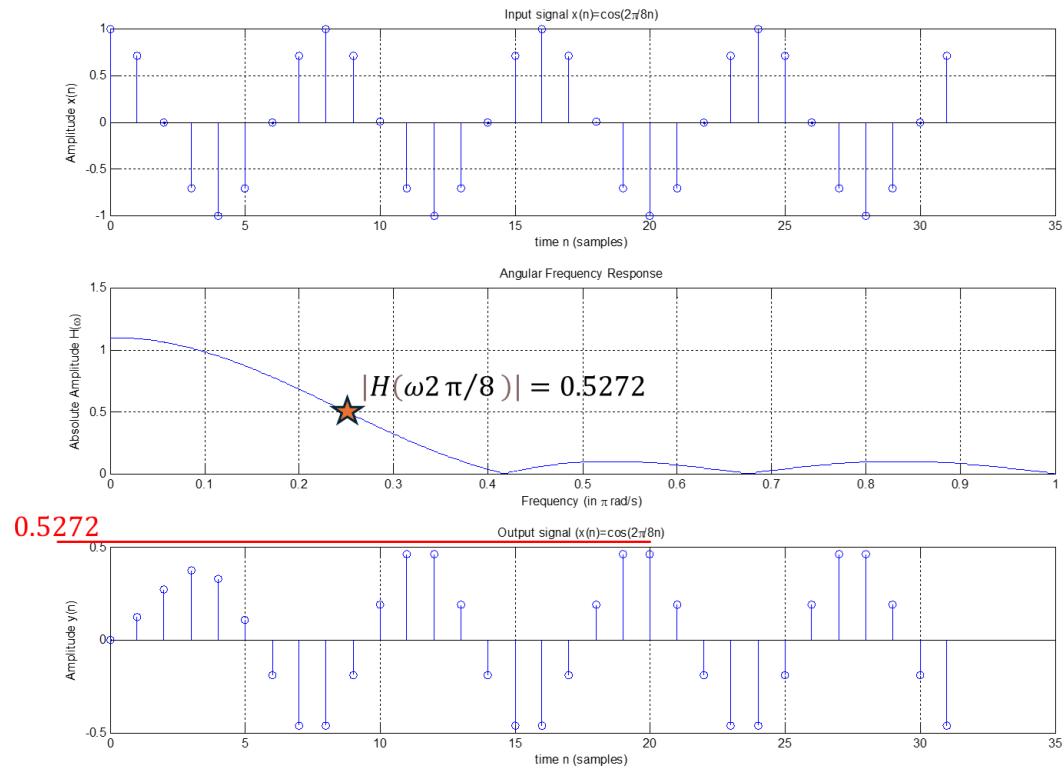
$$x(n) = \cos(\omega n) \xrightarrow{FIR} y(n) = |H(\omega)| \cos(\omega n - \varphi)$$

where $|H(\omega)|$ is the magnitude of the filter's frequency response, which depends on its impulse response $h(n)$. For the given impulse response and the circular frequencies $\omega_1 = 2\pi/16$ and $\omega_2 = 2\pi/8$, it holds that $|H(2\pi/16)| = 0.9264$ and $|H(2\pi/8)| = 0.5272$.

The next two figures show the input, the magnitude of the frequency response, and the output of the filter for the frequency cases $\omega_1 = 2\pi/16$ and $\omega_2 = 2\pi/8$. Note that in the first case ($\omega_1 = 2\pi/16$) the amplitude of the output signal $y(n)$ is approximately 0.9264, while in the second case ($\omega_2 = 2\pi/8$), the amplitude of the output signal $y(n)$ is approximately 0.5272. From the frequency response, it is evident that the filter is a lowpass filter with cut-off frequency $\pi/4$.

The above observations will help you verify the correct functionality of your programs.





Furthermore to use NEON the requirements are:

- Usage of directive `#include <arm_neon.h>` in the source code.
- Usage of directive `-mfpu=neon` in compiler (C/C++ Building Settings → ARM v7 gcc compiler → Miscellaneous → Other flags).

To use function `cos()` and the constant π (`M_PI`) the requirements are:

- Usage of directive `#include <math.h>` in the source code.
- Usage of directive `-lm` to linker (C/C++ Building Settings → ARM v7 gcc linker → Libraries → Other flags).

4. Group Formation - Assignment Submission

- Each working group may consist of up to 2 students. Fill in the details of the group members in the table on the first page of this assignment (name, registration number, and at least one contact email or phone number for each group).
- You must submit:
 - a. The source codes,
 - b. The results you obtain for each input signal case, and
 - c. The absolute time measurements for the two programs (serial and parallel) as well as the speedup and efficiency measurements.

Items (b) and (c) should be included in a separate Word document.

- Assignment submission date: **January 31st, 2025**.
- A date for the assignment evaluation will be scheduled.
- The assignment must be submitted electronically on the course website.